Conversions

1 Convert the following measurements:

- **a** $1\ \text{cm}$ to millimetres.
- **b** $3\ \text{cm}$ to millimetres.
- **c** $4.65\ \text{cm}$ to millimetres.
- **d** $60\ \text{mm}$ to centimetres.
- **e** $88\ \text{mm}$ to centimetres.
- **f** $1\ \text{m}$ to centimetres.
- **g** $6\ \text{m}$ to centimetres.
- **h** $7.33\ \text{m}$ to centimetres.
- **i** $800\ \text{cm}$ to metres.
- **j** $14.9\ \text{cm}$ to metres.
- **k** $13\,898\ \text{cm}$ to metres.
- **l** $9000\ \text{m}$ to kilometres.
- **m** $12\,226\ \text{m}$ to kilometres.
- **n** $1\ \text{km}$ to metres.
- **o** $6\ \text{km}$ to metres.
- **p** $6.69\ \text{km}$ to metres.

2 Convert the following measurements:

- **a** $1\ \text{m}$ to millimetres.
- **b** $14\ \text{m}$ to millimetres.
- **c** $9.03\ \text{m}$ to millimetres.
- **d** $3484\ \text{mm}$ to metres.
- **e** $6\ \text{km}$ to centimetres.
- **f** $7.43\ \text{km}$ to centimetres.
- **g** $866\,800\ \text{cm}$ to kilometres.
- **h** $9\ \text{km}$ to millimetres.
- **i** $6.71\ \text{km}$ to millimetres.
- **j** $1\,528\,400\ \text{mm}$ to kilometres.

Applications

3 A desk is $1.39\ \text{m}$ long. A rug is $120\ \text{cm}$ long.

- **a** Convert the length of the rug from centimetres to metres.
- **b** State whether the desk or the rug is longer.

4 Purnjay is $176\ \text{cm}$ tall. Nadia is $1.51\ \text{m}$ tall.

- **a** Convert Purnjay's height from centimetres to metres.
- **b** Which person is shorter?

5 State which person is taller in the following pairs:

- **a** Haonan is $165\ \text{cm}$ tall and Tina is $1.53\ \text{m}$ tall.
- **b** Hermione is $1.23\ \text{m}$ tall and Sandy is $161\ \text{cm}$ tall.
- **c** Joanne is $127\ \text{cm}$ tall and Ellie is $1.41\ \text{m}$ tall.

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6 A coffee table is $1.17\ \text{m}$ long. A whiteboard is $147\ \text{cm}$ long.

- **a** Convert the length of the coffee table from metres to centimetres.
- **b** State whether the coffee table or the whiteboard is longer.

7 Plant 1 grows $3.6\ \text{cm}$ in a week. Plant 2 grows $39\ \text{mm}$ in the same week.

Which plant grows more in the week?

8 A rectangular table is measured to be 586 centimetres long and 3.64 metres wide.

- **a** Convert the length of the rectangular table to metres.
- **b** Calculate the difference between the length and the width in metres.

9 A rectangular table is measured to be 5.23 metres long and 4.99 metres wide. Calculate the difference in metres between the length and the width of the table.

10 The Qinghai-Tibet railway in China includes the Tanggula railway station. At 5068 metres high, it is the world's highest railway station.

- **a** How many centimetres high is the station?
- **b** How many kilometres high is the station?

11 If a fly is $1.16\ \text{cm}$ long and a bee is $19\ \text{mm}$ long, how much longer is the bee than the fly in centimetres?

12 If a mosquito is $1.23\ \text{cm}$ long and a wasp is $13\ \text{mm}$ long, how much longer is the mosquito than the wasp in millimetres?

13 A tower is 125 metres tall.

- **a** Find the height of the building in kilometres.
- **b** Find the height of the building in centimetres.

14 The following table shows the distance from New York to other cities in the United States:

City	Distance (km)
Los Angeles	3940
Houston	2284
Washington	328
Chicago	1146
Charlotte	856

- **a** Which city is furthest from New York?
- **b** Which city is closest to New York?
- **c** Which city is about $1000\ \text{km}$ further away from New York than Charlotte?

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City	Distance (km)
New Orleans	1882

15 A car is 4 m long. A truck is 934 cm long.

a Find the length of the truck in metres.

b How many metres longer is the truck than the car?

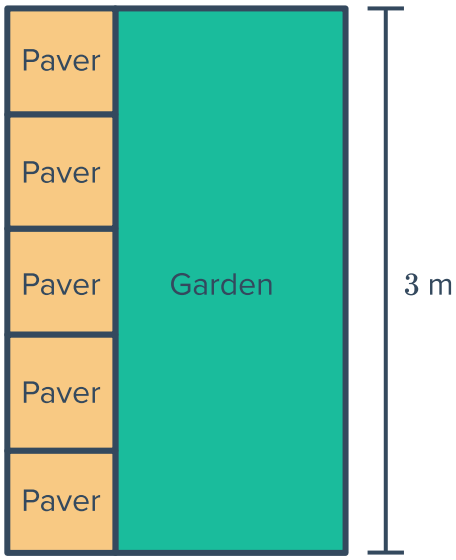
16 A sail boat is 6 m long. A yacht is 956 cm long.

a Find the length of the sail boat in centimetres.

b How many centimetres longer is the yacht than the sail boat?

17 Michael wants to lay a row of square pavers along the edge of his 3 m garden bed as shown in the picture:

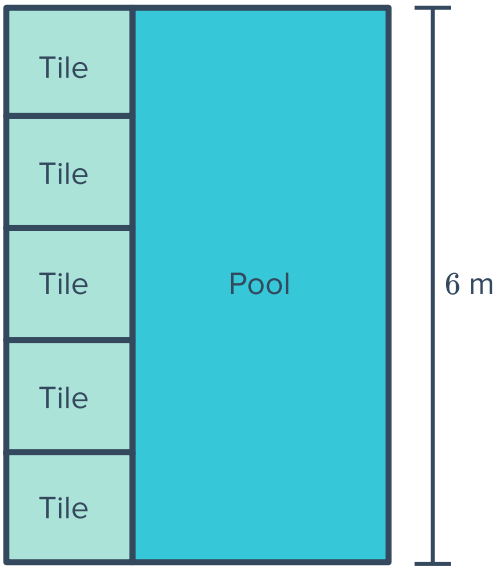
Find the side length of each paver in metres.



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18 Ryan wants to lay a row of square tiles along the edge of his 6 m pool as shown in the picture:

Find the side length of each tile in metres.



19 The table below shows the different skyscrapers around the world and their heights:

Name	Height (meters)	Height (kilometers)
One World Trade Center	541.3	
Shanghai Tower		0.632
Petronas Twin Towers	451.9	
Zifeng Tower		0.45
Burj Khalifa	828	

a Complete the conversions for the given skyscrapers in the table above.

b Which skyscraper is the tallest?

20 A brick is 230 mm long.

a Find the length of a single brick in centimetres.

b If 19 bricks were lined up lengthways to build a fence, how long would the fence be? Give your answer in centimetres.

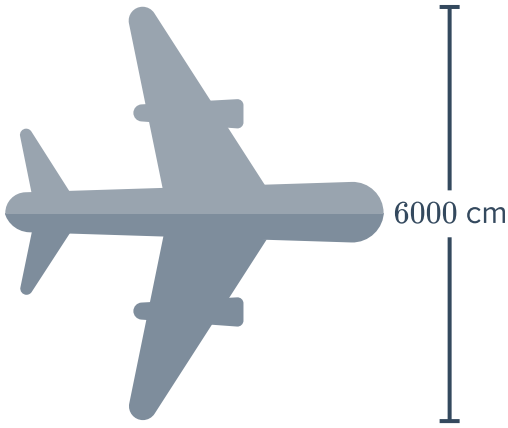
c Find the length of the fence in metres.

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21 The wingspan of a jumbo jet is 6000 cm:

a If 21 jumbo jets are lined up wing-to-wing, find the length of the total wingspan in centimetres.

b Convert this length to kilometres.



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